

JAVA PROGRAMMING LABORATORY

Experiment No: 05

Date: 21/09/2026

Title: Demonstration of Iterator Interface Methods in Java

1. Objective / Aim

To understand and demonstrate the Iterator interface in Java for traversing elements of a collection using hasNext(), next(), remove(), and forEachRemaining() methods.

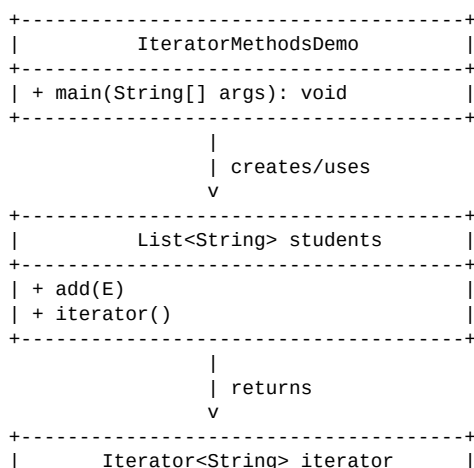
2. System Requirements

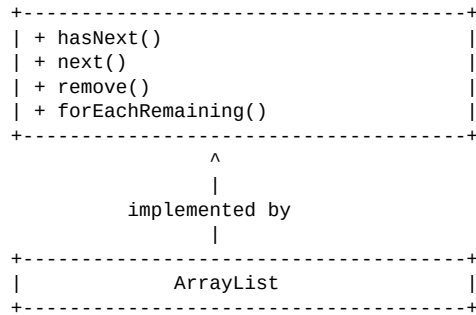
Hardware Requirements	Software Requirements
Computer/Laptop	Operating System: Windows / Linux / macOS
Minimum 4 GB RAM	Java Development Kit (JDK) 17 or above
Minimum 1 GHz processor	IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
At least 100 MB free storage	Java compiler and Command Prompt/Terminal

3. Algorithm / Logic Description

1. Import ArrayList, Iterator, and List from java.util.
2. Create a List of String elements using ArrayList.
3. Add five student names to the list.
4. Display the original list.
5. Create an Iterator using the list's iterator() method.
6. Use hasNext() to check whether another element is available.
7. Use next() to retrieve and display each element.
8. If the current student is Arjun, use iterator.remove() to remove that element safely.
9. Display the list after removal.
10. Create a new Iterator and use next() to move to the first element.
11. Use forEachRemaining() to process and display all remaining elements.

4. UML Class Diagram





5. Source Code

```

import java.util.ArrayList;
import java.util.Iterator;
import java.util.List;

class IteratorMethodsDemo {

    public static void main(String[] args) {

        // Creating a List
        List<String> students = new ArrayList<>();

        students.add("Rahul");
        students.add("Priya");
        students.add("Arjun");
        students.add("Meera");
        students.add("Anjali");

        System.out.println("Original List: " + students);

        // Creating Iterator
        Iterator<String> iterator = students.iterator();

        // hasNext() and next()
        System.out.println("\nUsing hasNext() and next():");

        while (iterator.hasNext()) {

            String student = iterator.next();

            System.out.println("Student: " + student);

            // remove()
            if (student.equals("Arjun")) {
                iterator.remove();
                System.out.println("Arjun removed.");
            }
        }

        System.out.println("\nList after remove(): " + students);

        // Creating a new Iterator
        iterator = students.iterator();

        // Moving to the first element
        if (iterator.hasNext()) {
            System.out.println("\nFirst element: " + iterator.next());
        }

        // forEachRemaining()
        System.out.println("\nRemaining elements using forEachRemaining():");

        iterator.forEachRemaining(student -> {
            System.out.println("Student: " + student);
        });
    }
}

```

6. Completion and Execution Commands

Step 1: Save the program as:
IteratorMethodsDemo.java

Step 2: Compile the program:
javac IteratorMethodsDemo.java

Step 3: Execute the program:
java IteratorMethodsDemo

7. Output

Original List: [Rahul, Priya, Arjun, Meera, Anjali]

Using hasNext() and next():

Student: Rahul

Student: Priya

Student: Arjun

Arjun removed.

Student: Meera

Student: Anjali

List after remove(): [Rahul, Priya, Meera, Anjali]

First element: Rahul

Remaining elements using forEachRemaining():

Student: Priya

Student: Meera

Student: Anjali

8. Result / Completion

The program was successfully compiled and executed. The Iterator interface methods hasNext(), next(), remove(), and forEachRemaining() were demonstrated for traversing and modifying an ArrayList.